

Automated Network Request Management Using Servicenow(ANRM)

Project Documentation

Project Title

**Automated Network Request Management (ANRM) Using
ServiceNow**

Objective

The objective of the Automated Network Request Management (ANRM) project is to automate the process of handling network-related service requests within an organization. The application enables users to submit network requests, allows authorized approvers to review and approve or reject requests, sends automated email notifications, provides reporting capabilities, and supports self-service request status checking.

Project Overview

The ANRM system is a fully automated ServiceNow application that allows users to request network access services such as VPN, admin access, or standard access.

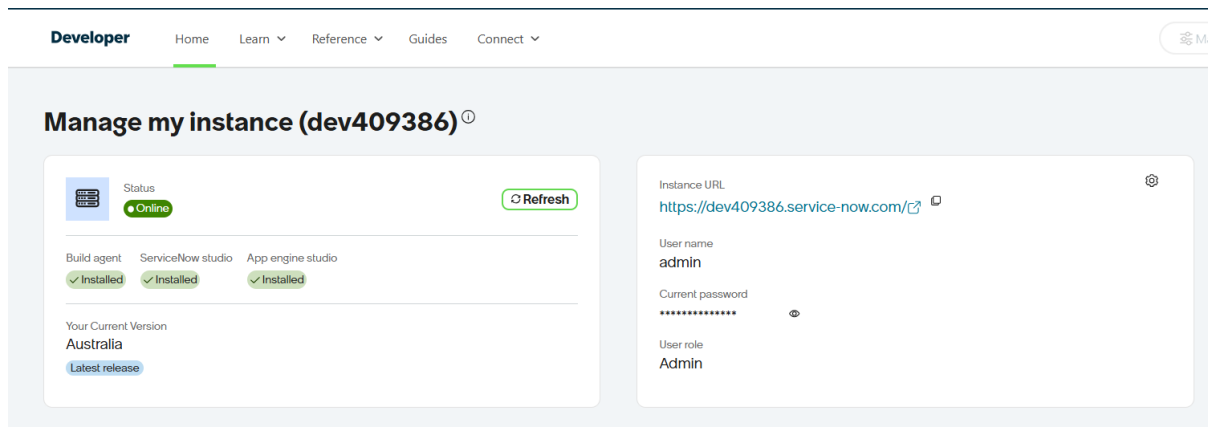
The system includes:

- Service Catalog for request submission
- Flow Designer for automation
- ACL for security
- Notifications for communication

- Dashboard for monitoring

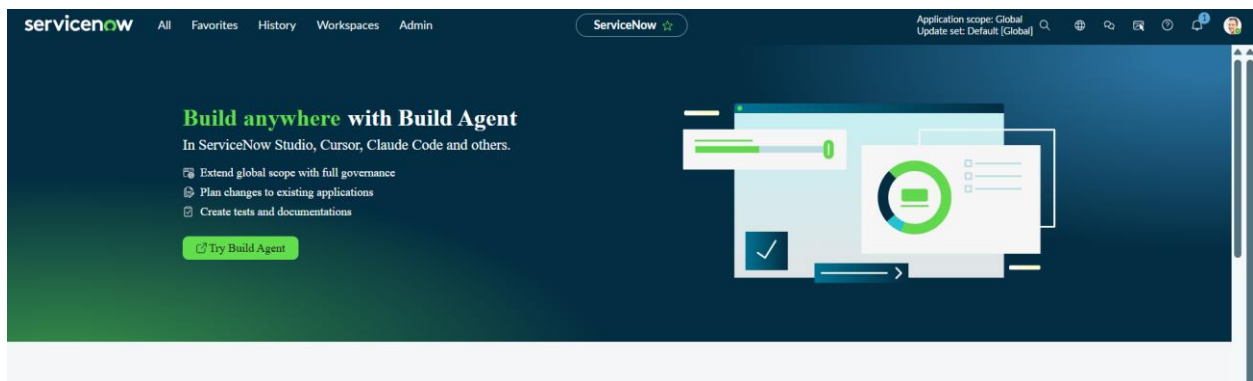
Step 1: Create Personal Developer Instance (PDI)

1. Visit developer.servicenow.com.
2. Sign in or create a developer account.
3. Request a Personal Developer Instance.
4. Wait for instance activation.
5. Login using admin credentials.



Step 2: Create Scoped Application

Navigation:



Step 3: Configure User Roles and Groups

Navigation:

User Administration → Roles / Groups

- Created roles for requester, approver, engineer
- Assigned roles to respective groups
- Added users into groups

Groups

Name

Search

Step 4: Create Catalog Item

Navigation:

Service Catalog → Maintain Items → New

Catalog Item Name: Network Access Request

Variables:

- Request Type
- Access Level
- Device
- Requested For
- Business Justification
- VPN Details
- IP Address
- Decommission Reason

Save and modify items based on the improved [submitting an item](#)

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name

Application

Catalogs

Category

State

Checked out

Owner

Secondary owners

Active ☒

Fulfillment automation level

Item Details Process Engine **Picture** Pricing Portal Settings

Add an icon and picture for display

Icon [Click to add...](#)

Picture [Click to add...](#)

[Submit](#) [Try It](#)

Step 5: Client Script Implementation

Type: onChange

Field: Request Type

Function:

- Show VPN field when VPN selected
- Show IP field when new connection selected
- Hide unnecessary fields

Client Script New record

New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name

Table

UI Type

Type

Field name

Application

Active ☒

Inherited ☐

Global ☒

Description

Messages

Script

```

1 function onChange(control, oldValue, newValue, isLoading, isTemplate) {
2   if (isLoading || newValue === '') {
3     return;
4   }
5   //Type appropriate comment here, and begin script below
6
7
8 }

```

Isolate script ☐

[Submit](#)

Step6: UI Policies

- VPN → Show VPN Details

- New Connection → Show IP Address
- Decommission → Show Reason

ServiceCatalog > Top Requests > Network Access Request

Request network access - VPN, Firewall, WiFi, or Server Access

Exists in categories

* Request Type
VPN

* Access Level
Read

* Device Type
Laptop

* Business Justification

* Priority
Low

Order this Item
Quantity: 1
Delivery time: 0 Days
Order Now
Add to Cart
Shopping Cart: Empty

Step 6: Access Control (ACL)

Operation	Access
Create	Requester
Read	All roles
Write	Approver, Engineer
Delete	Approver

Access Controls for text Search				
Name	Decision Type	Operation	Type	
sys_auth_mfa_context_mfa_factor_m2m	Allow If	read	record	
3e81e788f072210755cfffffffc8	Deny Unless	execute	ux_data_broker	
sys_notification_actionable_content	Allow If	delete	record	
sn_bm_client_recommendation_eval	Allow If	report_on	record	
ecc_action	Deny Unless	read	record	
sys_m2m_email_report_attachment	Deny Unless	delete	record	
sys_pd_activity_override	Deny Unless	create	record	
cmdb_rel_filter	Deny Unless	report_view	record	
578dedff53622110b87eddeeff7b1247	Allow If	execute	ux_data_broker	
sys_app_template_output_var.type	Allow If	write	record	

Step 7: Flow Designer Implementation

Trigger:

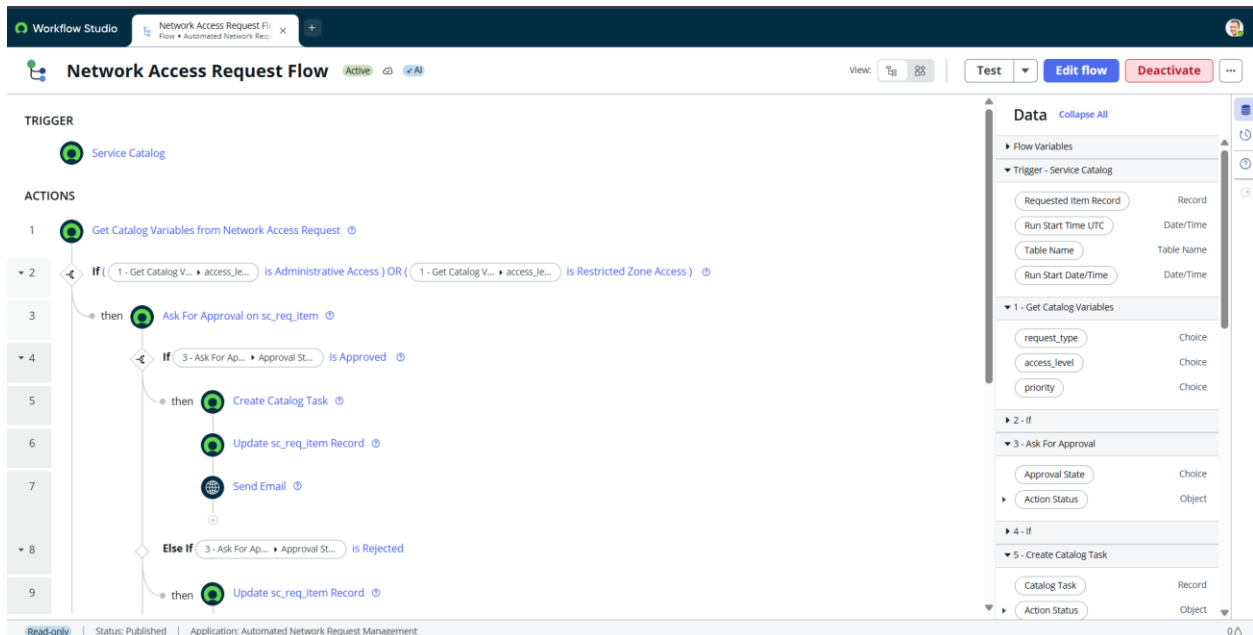
- Catalog Item Request Submitted

Logic:

- If Access Level = Admin → Approval Required
- Else → Auto Approved

Actions:

- Create approval task
- Send notification
- Create fulfillment task
- Assign to Network Team
- Update request status



Step 8: Email Notifications

Created 3 notifications:

1. Request Submitted → Sent to Approver
2. Status Updated → Sent to Requester
3. Request Completed → Sent to Requester

Preview Email

Hello System Administrator,

Unfortunately, your network access request has been **approval**.

Request Number: RITM0010001

Description: Request network access - VPN, Firewall, WiFi, or Server Access

Comments:

If you believe this was made in error or would like to resubmit with additional justification, please contact your manager or the Network Approvers team.

[View your request](#)

Thank you.

Ref:MSG0000196_6UFsesmZqu0hWdy2681x

Step 9: Dashboard Creation

Dashboard URL:

x_1877505_automa_1_dashboard.do

Dashboard Displays:

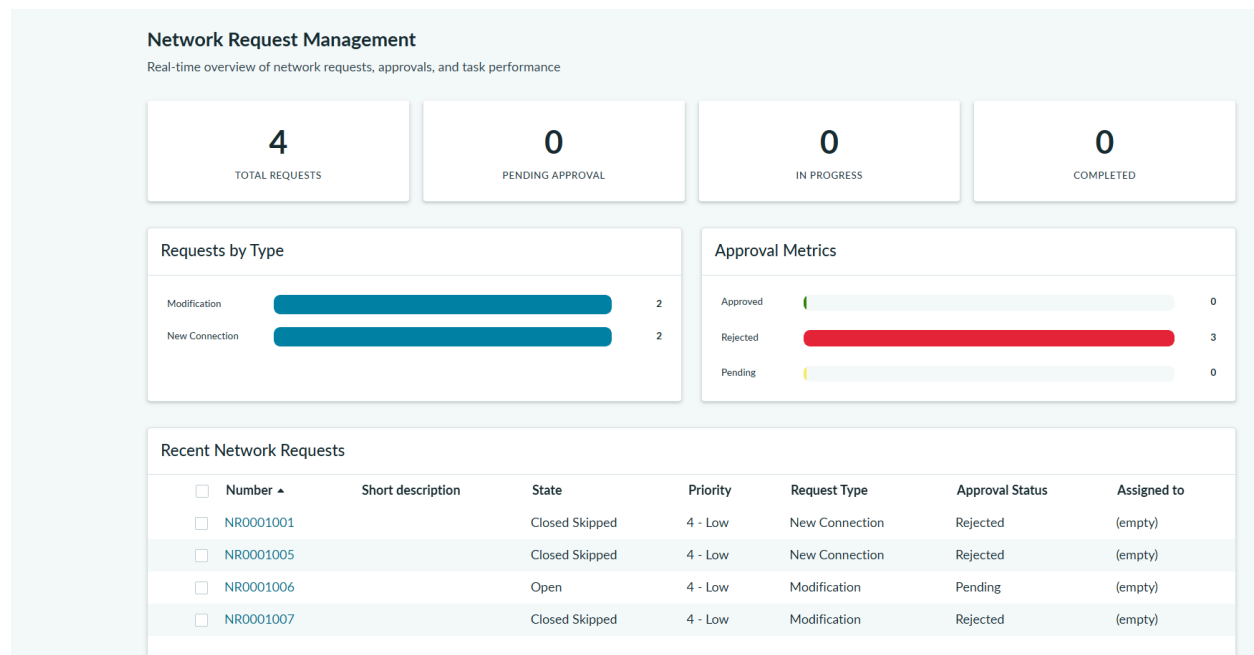
- Total Requests
- Pending Requests
- In Progress Requests
- Completed Requests

Step 9: Reports

Report Type: Bar Chart

Table: Network Request

Group By: Request Type



Learning Outcomes

- Created notifications.
- Verified email processing.
- Configured list layouts.
- Developed reports.

Final Testing Checklist

- ✓ Scoped Application
- ✓ Network Request Table
- ✓ Network Asset Table
- ✓ Modules
- ✓ UI Policy
- ✓ ACL

- ✓ Custom Role
- ✓ Business Rules
- ✓ Email Notifications
- ✓ Reports
- ✓ AI Component
- ✓ End-to-End Testing
- ✓ Application Export

Conclusion

The Automated Network Request Management (ANRM) application successfully automates the lifecycle of network service requests using the ServiceNow platform. The project demonstrates CSA and CAD concepts including scoped application development, custom tables, business rules, access controls, notifications, reporting, and AI-assisted request querying. The solution improves operational efficiency, enhances security through role-based access, and provides better visibility into network request processing.

Future Enhancements

- Multi-level approvals.
- Network asset inventory dashboard.
- SLA tracking.
- Integration with CMDB.
- Service Portal interface.
- Now Assist integration.
- Mobile application support.

- Advanced analytics dashboard.
- API integration with network monitoring tools.